

### Example 6.1

$$X_t = \phi X_{t-1} + Z_t \quad |\phi| < 1 \text{ (weakly stationary)}$$

For the AR(1) process

$$\rho(h) = \phi^{|h|} \text{ (Lecture 3).}$$

Partial autocorrelation function (PACF)

At lag 1

$$\rho(0,1) = \rho(1) = \phi.$$

$$\begin{aligned} \rho(0,2|1) &= \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} \\ &= \frac{\phi^2 - \phi^2}{1 - \phi^2} = 0. \end{aligned}$$

$$\rho(0,3|1,2) = \frac{\rho_3 - \rho_2 \beta_{2,1}}{1 - \phi \beta_{2,1}}$$

$$\beta_{2,1} = \frac{\rho_1(1 - \rho_2)}{1 - \rho_1^2} = \frac{\phi(1 - \phi^2)}{1 - \phi^2} = \phi$$

$$\rho(0,3|1,2) = \frac{\phi^3 - \phi^3}{1 - \phi^2} = 0$$